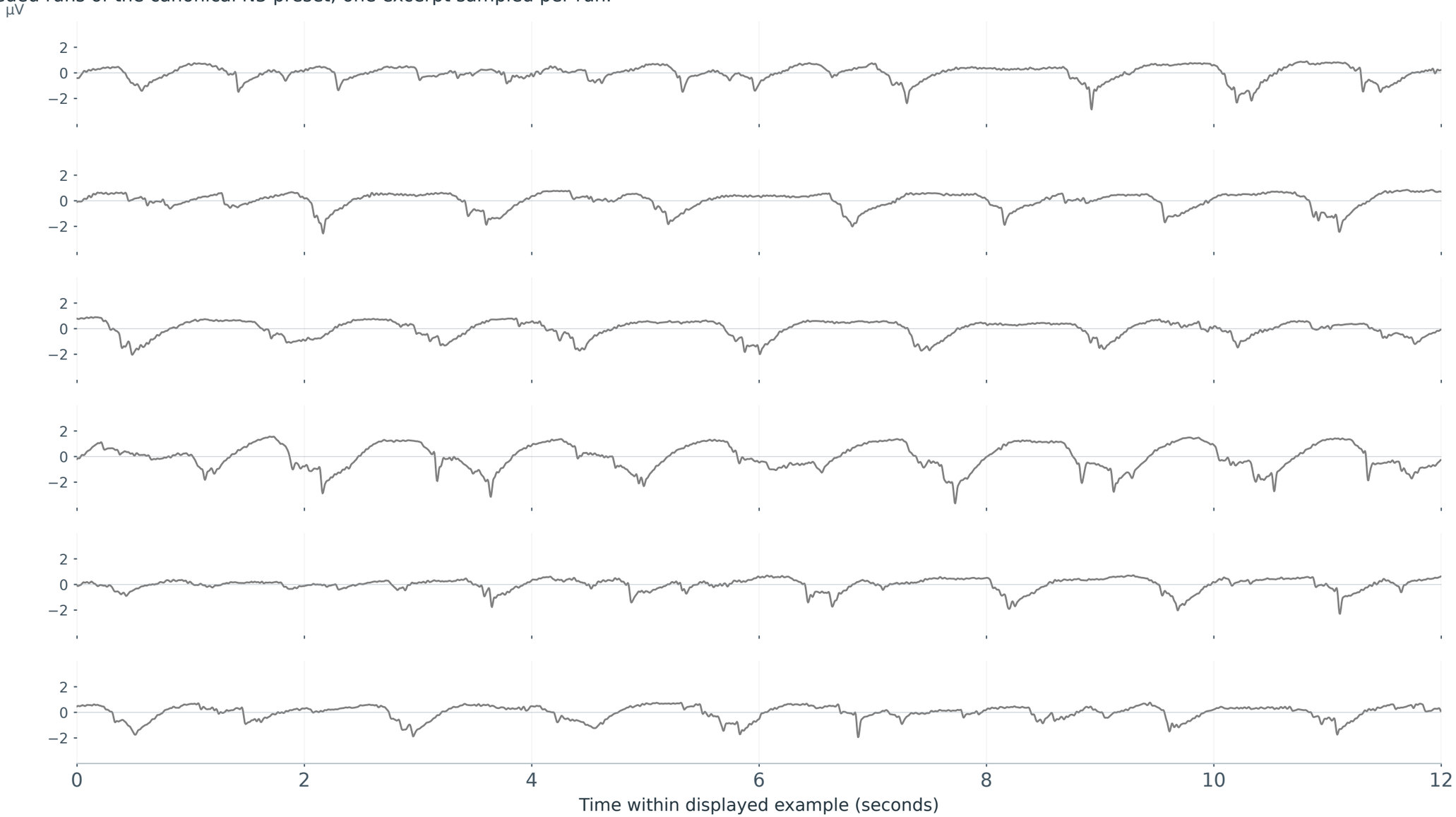
Fink N3 examples

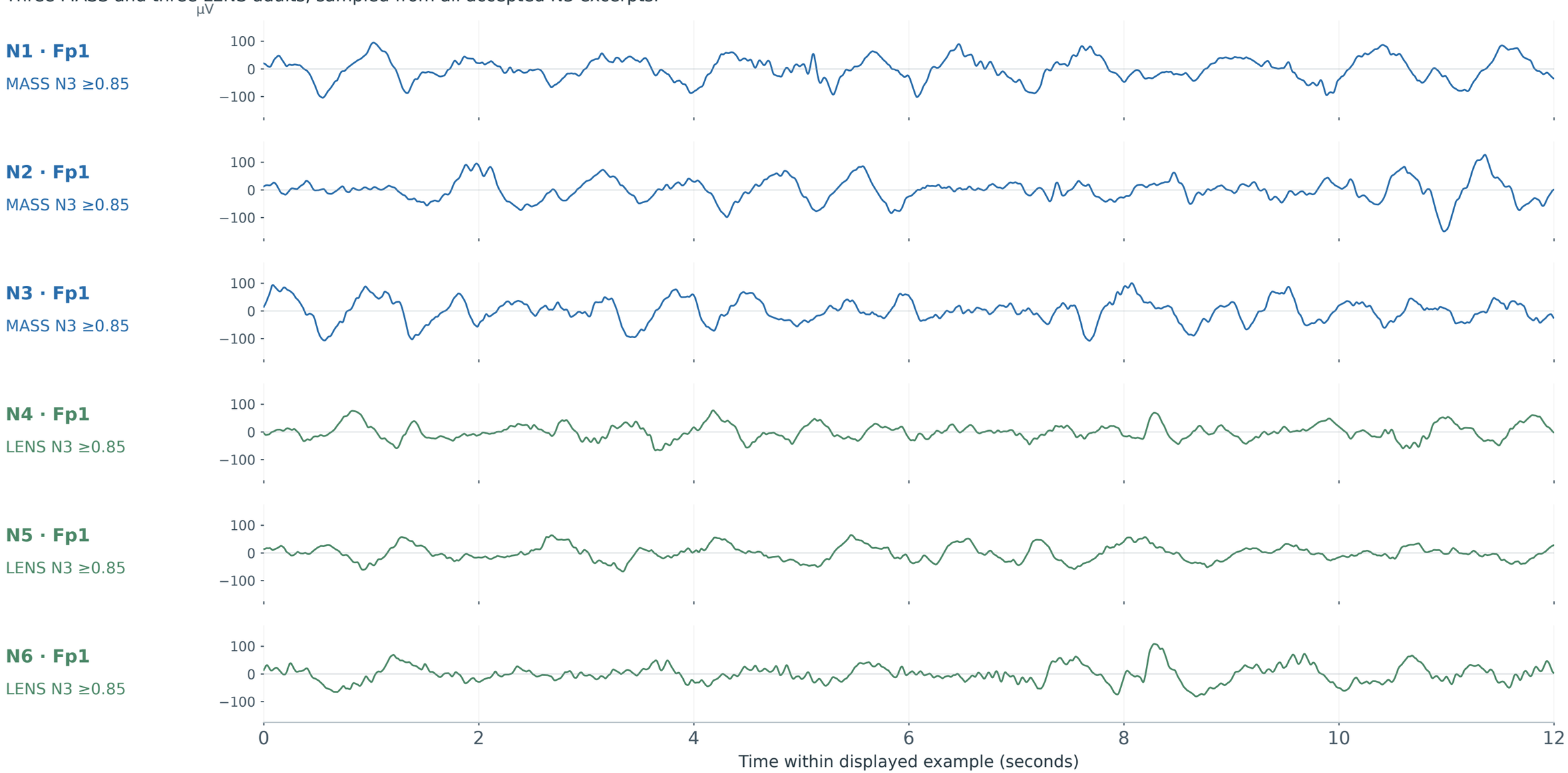
Six independently seeded runs of the canonical N3 preset; one excerpt sampled per run.



12 seconds shown; 0.5–30 Hz local LFP. Model amplitude has its own scale and no established scalp-voltage conversion.

N3 examples

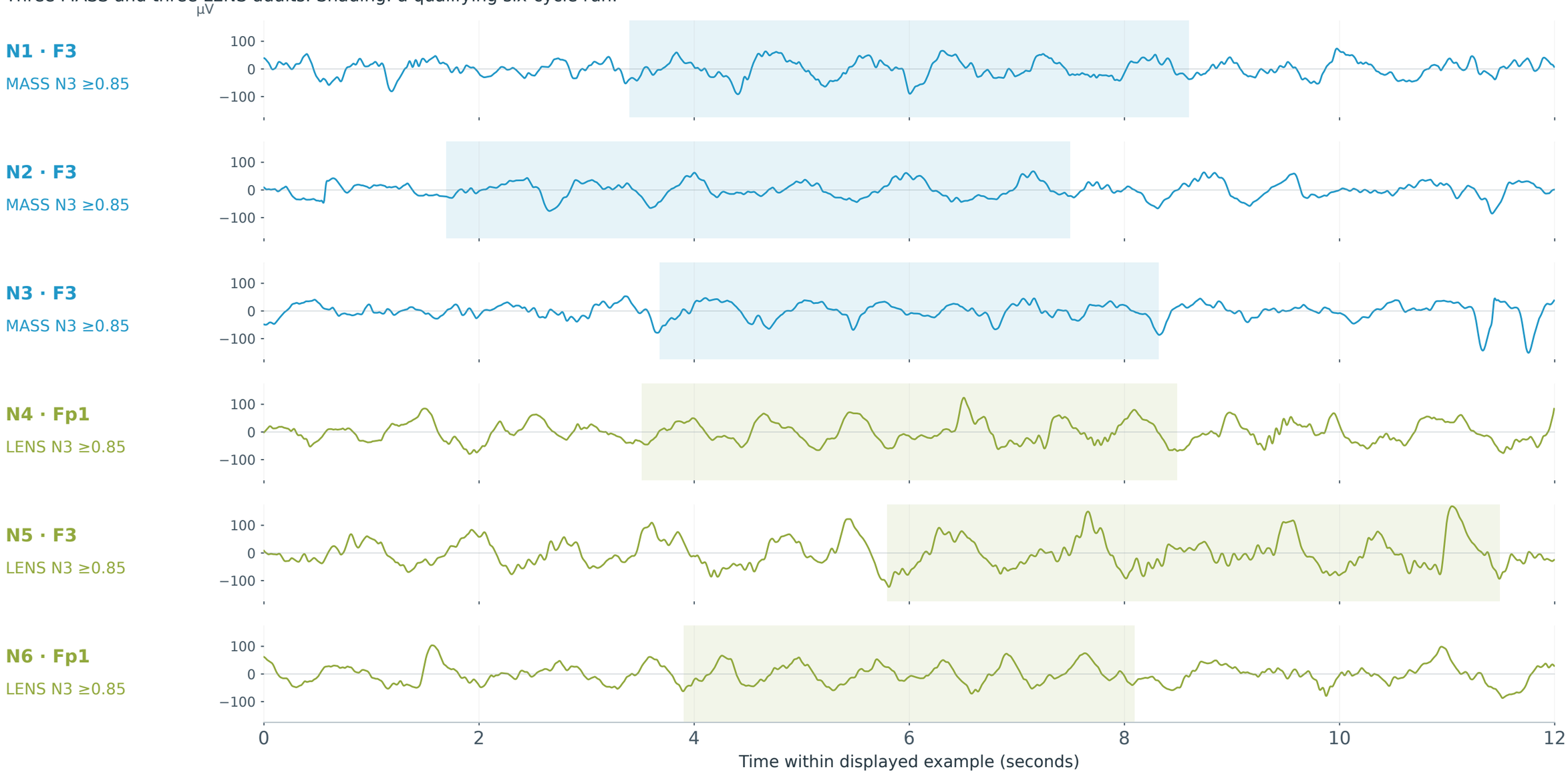
Three MASS and three LENS adults, sampled from all accepted N3 excerpts.



12 seconds shown; 0.5–30 Hz common-average EEG. Same voltage scale as the cyclic-N3 examples.

Cyclic N3 examples

Three MASS and three LENS adults. Shading: a qualifying six-cycle run.



12 seconds shown; 0.5–30 Hz common-average EEG. Same voltage scale as the N3 examples.

Human N3 versus Fink N3

Cutoff / model	20 s excerpts	Sessions / model runs	Adults
MASS N3 ≥0.85	21	8	8
LENS N3 ≥0.85	332	10	9
MASS cyclic N3 ≥0.85	6	4	4
LENS cyclic N3 ≥0.85	112	7	6
Fink N3 LFP	225	25	-

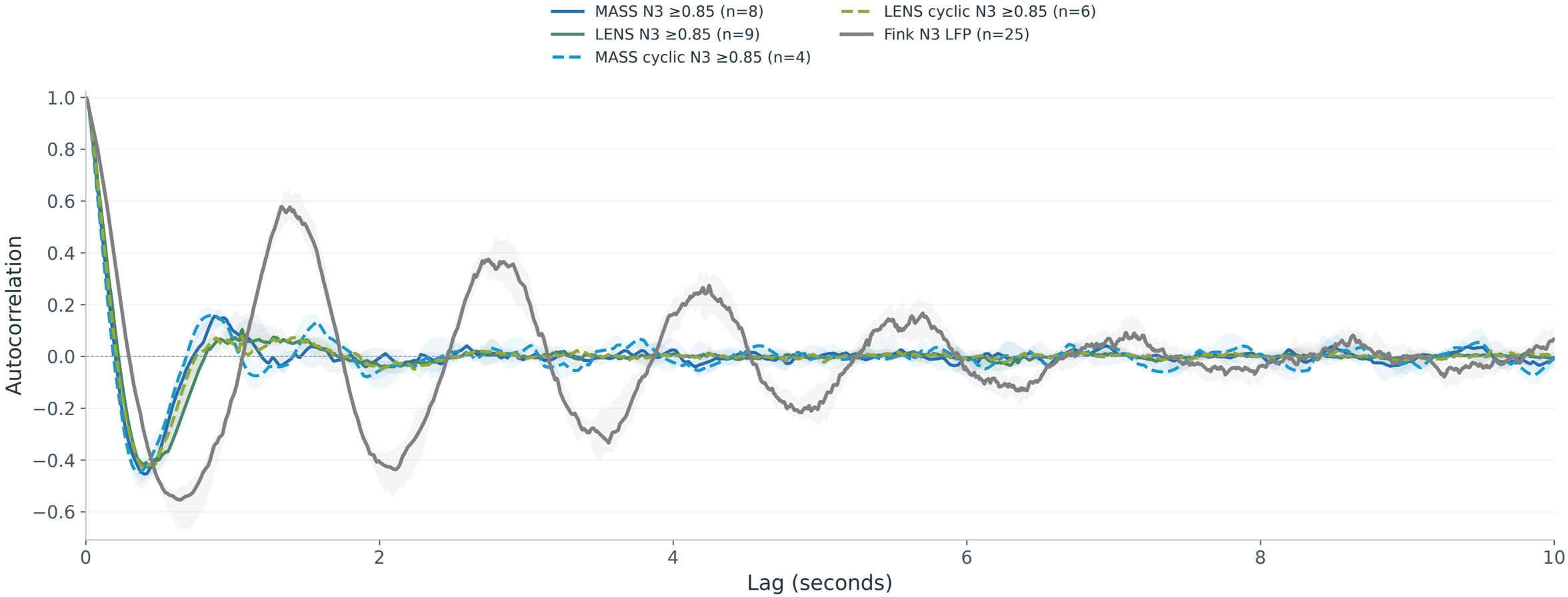
All human MASS groups also require expert N3. Human figures combine the nine measured electrodes.

Cyclic N3: six consecutive cycles ≥75 µV; duration SD / mean ≤0.20; overlapping left/right runs.

Original N3 includes cyclic excerpts. No statistical test compares original N3 with its cyclic subset.

Stars compare Fink with each of the four human N3 groups.

Autocorrelation



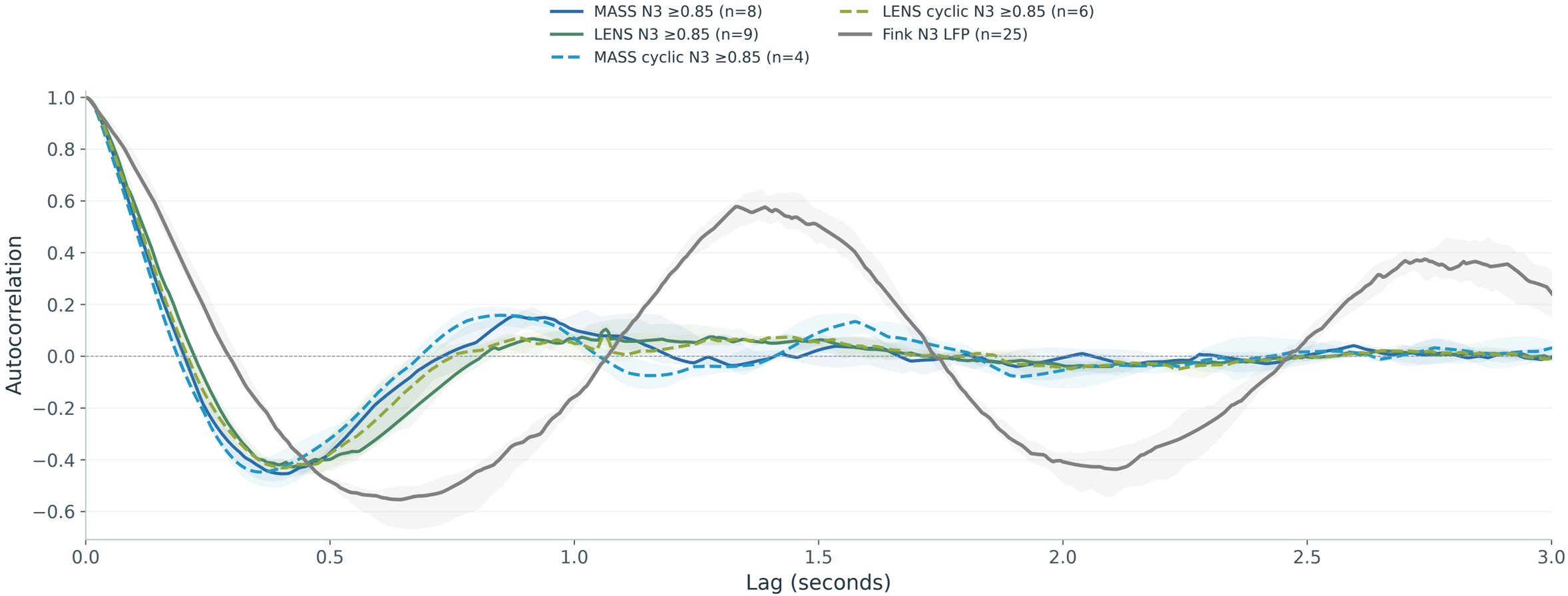
Lines: median; shading: middle 50%. One observation per adult or model run.

Whole-curve test: ACF 0-10 s; both ACF views use the same test.

Comparison	MASS N3 ≥0.85	LENS N3 ≥0.85	MASS cyclic N3 ≥0.85	LENS cyclic N3 ≥0.85
vs Fink N3 LFP	*	*	*	*

Holm-adjusted: * <0.05 ** <0.01 *** <0.001 **** <0.0001 ns ≥0.05 NA unavailable

Autocorrelation - first 3 seconds



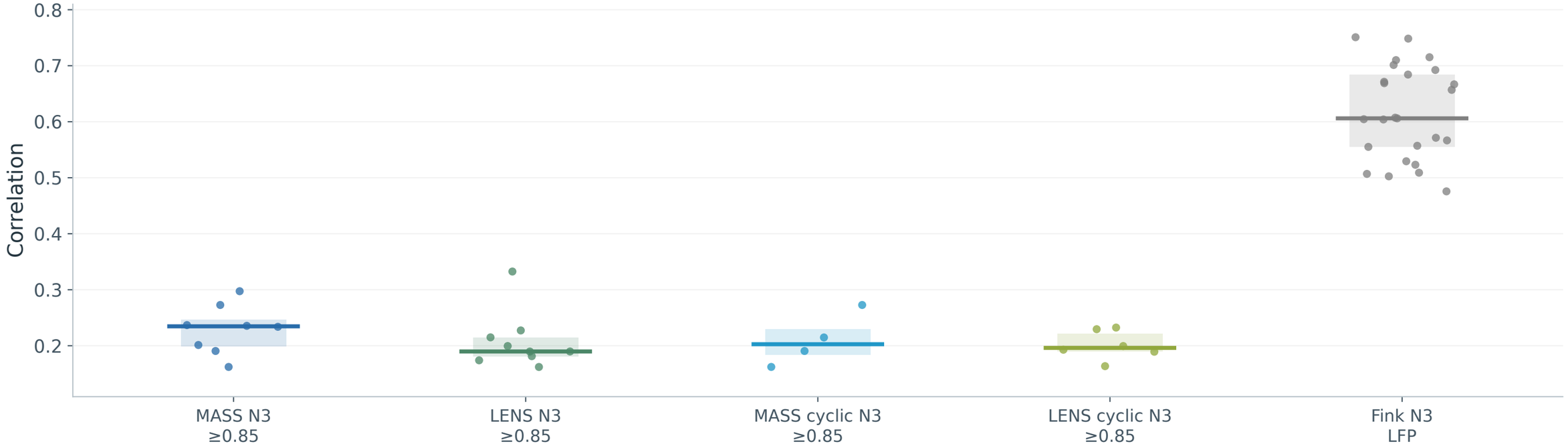
Lines: median; shading: middle 50%. One observation per adult or model run.

Whole-curve test: ACF 0-10 s; both ACF views use the same test.

Comparison	MASS N3 ≥0.85	LENS N3 ≥0.85	MASS cyclic N3 ≥0.85	LENS cyclic N3 ≥0.85
vs Fink N3 LFP	*	*	*	*

Holm-adjusted: * <0.05 ** <0.01 *** <0.001 **** <0.0001 ns ≥0.05 NA unavailable

Autocorrelation peak height



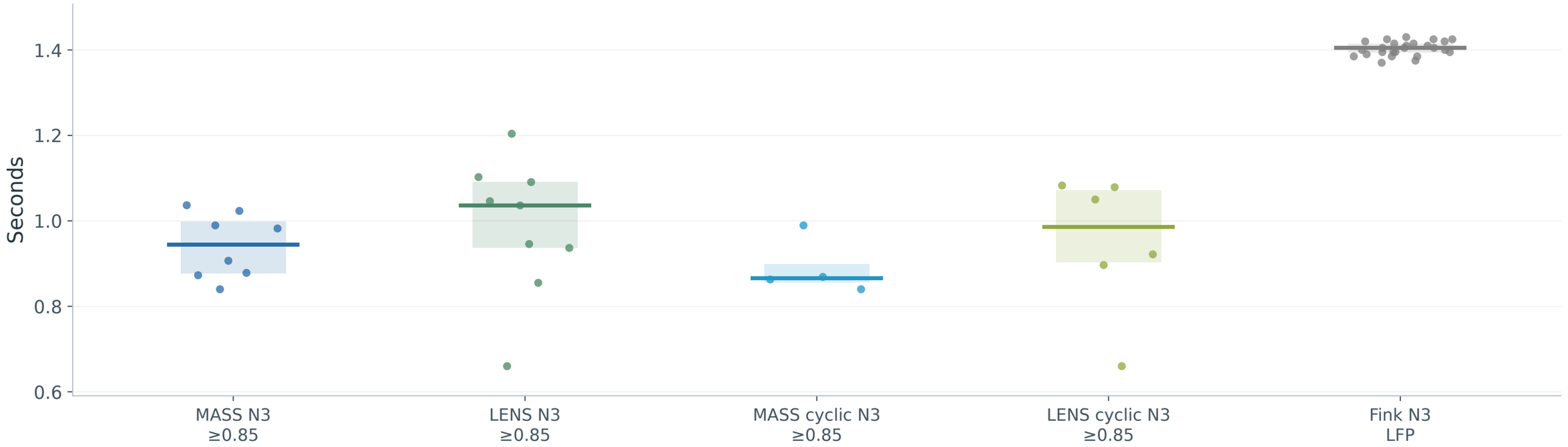
One dot per adult or model run; line: median; box: middle 50%.

Height of the first positive ACF peak after a zero crossing; higher values indicate stronger repetition.

Measurement	MASS N3 ≥0.85	LENS N3 ≥0.85	MASS cyclic N3 ≥0.85	LENS cyclic N3 ≥0.85	Fink N3 LFP
Median [middle 50%]	0.235 [0.199, 0.246]	0.19 [0.181, 0.215]	0.203 [0.184, 0.229]	0.196 [0.19, 0.222]	0.606 [0.555, 0.684]
n (adults / runs)	8	9	4	6	25
vs Fink N3 LFP	***	**	ns	**	-

Holm-adjusted: * <0.05 ** <0.01 *** <0.001 **** <0.0001 ns ≥0.05 NA unavailable

Autocorrelation peak lag



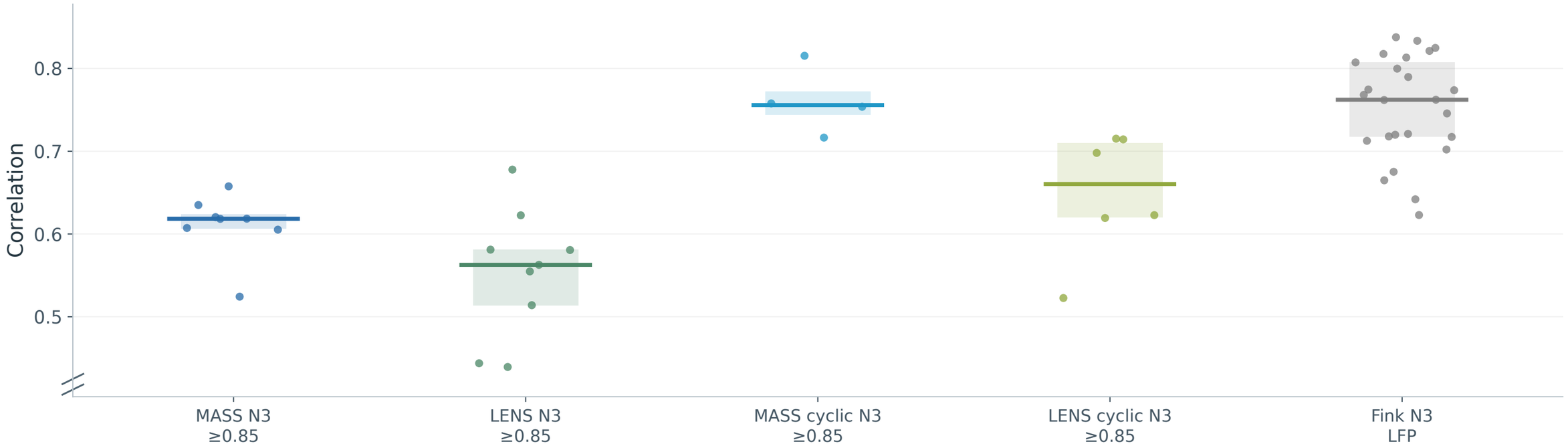
One dot per adult or model run; line: median; box: middle 50%.

Lag to the first positive ACF peak (0.25-2 s), approximately one repetition period.

Measurement	MASS N3 ≥0.85	LENS N3 ≥0.85	MASS cyclic N3 ≥0.85	LENS cyclic N3 ≥0.85	Fink N3 LFP
Median [middle 50%]	0.944 [0.877, 0.998]	1.04 [0.937, 1.09]	0.866 [0.857, 0.899]	0.986 [0.903, 1.07]	1.41 [1.4, 1.42]
n (adults / runs)	8	9	4	6	25
vs Fink N3 LFP	*	*	ns	*	-

Holm-adjusted: * <0.05 ** <0.01 *** <0.001 **** <0.0001 ns ≥0.05 NA unavailable

Successive-cycle correlation



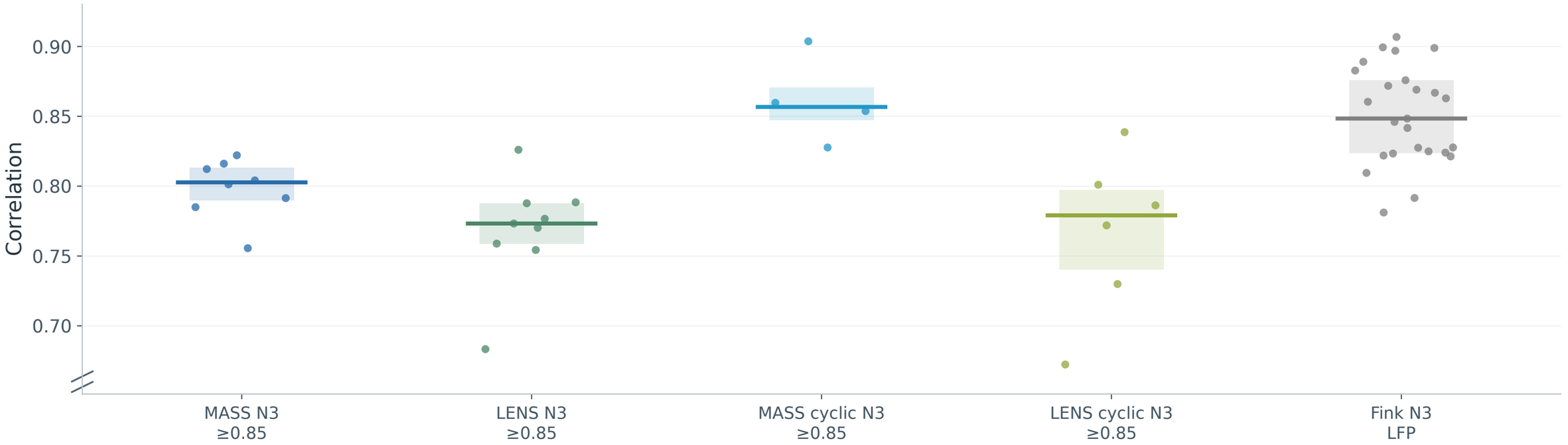
One dot per adult or model run; line: median; box: middle 50%.

Original neighboring retained cycles. Correlation range: -1 to +1.

Measurement	MASS N3 ≥0.85	LENS N3 ≥0.85	MASS cyclic N3 ≥0.85	LENS cyclic N3 ≥0.85	Fink N3 LFP
Median [middle 50%]	0.618 [0.607, 0.624]	0.563 [0.514, 0.581]	0.756 [0.744, 0.772]	0.66 [0.62, 0.71]	0.762 [0.717, 0.807]
n (adults / runs)	8	9	4	6	25
vs Fink N3 LFP	***	*	ns	ns	-

Holm-adjusted: * <0.05 ** <0.01 *** <0.001 **** <0.0001 ns ≥0.05 NA unavailable

Cycle-template correlation



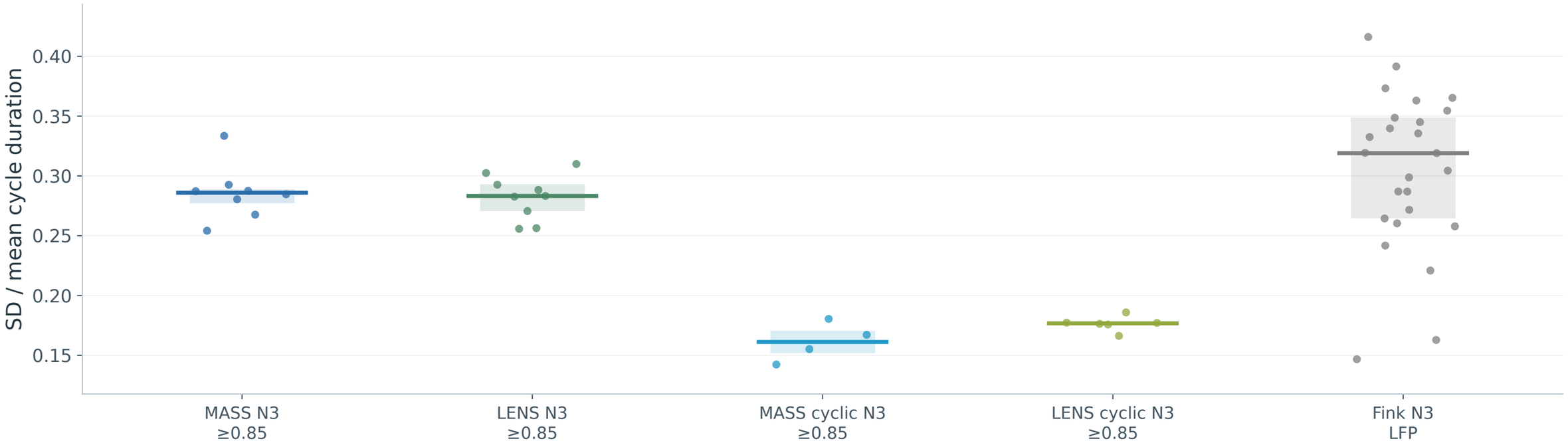
One dot per adult or model run; line: median; box: middle 50%.

Retained cycles compared with a leave-one-out median template. Correlation range: -1 to +1.

Measurement	MASS N3 ≥0.85	LENS N3 ≥0.85	MASS cyclic N3 ≥0.85	LENS cyclic N3 ≥0.85	Fink N3 LFP
Median [middle 50%]	0.803 [0.79, 0.813]	0.773 [0.759, 0.788]	0.857 [0.847, 0.871]	0.779 [0.74, 0.797]	0.848 [0.824, 0.876]
n (adults / runs)	8	9	4	6	25
vs Fink N3 LFP	ns	*	ns	ns	-

Holm-adjusted: * <0.05 ** <0.01 *** <0.001 **** <0.0001 ns ≥0.05 NA unavailable

Cycle-period variability



One dot per adult or model run; line: median; box: middle 50%.

Cycle-duration SD / mean. Cyclic N3 uses qualifying runs; pooled duration CV can exceed 0.20.

Measurement	MASS N3 ≥0.85	LENS N3 ≥0.85	MASS cyclic N3 ≥0.85	LENS cyclic N3 ≥0.85	Fink N3 LFP
Median [middle 50%]	0.286 [0.277, 0.289]	0.283 [0.271, 0.293]	0.161 [0.152, 0.171]	0.177 [0.176, 0.177]	0.319 [0.264, 0.349]
n (adults / runs)	8	9	4	6	25
vs Fink N3 LFP	ns	ns	ns	ns	-

Holm-adjusted: * <0.05 ** <0.01 *** <0.001 **** <0.0001 ns ≥0.05 NA unavailable

Numerical results

Measure	MASS N3 ≥0.85	LENS N3 ≥0.85	MASS cyclic N3 ≥0.85	LENS cyclic N3 ≥0.85	Fink N3 LFP
Autocorrelation peak height Correlation	0.235 [0.199, 0.246] n=8	0.19 [0.181, 0.215] n=9	0.203 [0.184, 0.229] n=4	0.196 [0.19, 0.222] n=6	0.606 [0.555, 0.684] n=25
Autocorrelation peak lag Seconds	0.944 [0.877, 0.998] n=8	1.04 [0.937, 1.09] n=9	0.866 [0.857, 0.899] n=4	0.986 [0.903, 1.07] n=6	1.41 [1.4, 1.42] n=25
Successive-cycle correlation Correlation	0.618 [0.607, 0.624] n=8	0.563 [0.514, 0.581] n=9	0.756 [0.744, 0.772] n=4	0.66 [0.62, 0.71] n=6	0.762 [0.717, 0.807] n=25
Cycle-template correlation Correlation	0.803 [0.79, 0.813] n=8	0.773 [0.759, 0.788] n=9	0.857 [0.847, 0.871] n=4	0.779 [0.74, 0.797] n=6	0.848 [0.824, 0.876] n=25
Cycle-period variability SD / mean cycle duration	0.286 [0.277, 0.289] n=8	0.283 [0.271, 0.293] n=9	0.161 [0.152, 0.171] n=4	0.177 [0.176, 0.177] n=6	0.319 [0.264, 0.349] n=25

Median [25th-75th percentile]. n = adults, or model runs for Fink. Available n varies by measurement.

Human N3 cycles ≥75 μV; cyclic N3 cycles also belong to qualifying runs. Fink cycles have no amplitude cutoff.

Reading the comparisons

Numerical results

Median and middle 50% across adults; Fink uses independently seeded model runs.

Comparison set

Fink versus each human N3 group.

Significance symbols

Two-sided Mann-Whitney U tests for scalar measures; adult/run-label permutation tests for whole curves.

Consistent adjustment

Existing Holm-adjusted values retain their original correction family. Repeated comparisons use the same stars.

Filtering

ACF and wave shapes: 0.5-30 Hz. Trough detection: 0.5-4 Hz.

Cycle interpretation

N3 cycle measurements require ≥75 µV. Cyclic N3 additionally requires regular timing.

Fink amplitude

Local model LFP has no established scalp-voltage conversion. Its cycles have no voltage cutoff.

Holm-adjusted: * <0.05 ** <0.01 *** <0.001 **** <0.0001 ns ≥0.05 NA unavailable